

Teamwork Agreement Template

For the MathWorks Classroom Challenge Projects

Project: Drone Payload Capacity and Structural Design Analysis

Team Name: banjo

Instructor: Dr. Kevin Graham

Program/Class: Engineering Pathway Program (EPP)

Members and Roles

1. Nathan Huynh - Documentation and Visualization Lead
2. Annie Dang - PM
3. Jonah Williamson - Analysis/Validation Lead
4. Bao Nguyen - Modeling Lead

1) Purpose & Scope

We agree to collaborate to deliver a high-quality, ethical, and reproducible project. Our work will follow the project guidance and produce the specified artifacts (e.g. plots, code, and visualizations).

2) Roles & Responsibilities

- Project Manager (PM): Schedules meetings; maintains timeline; ensures tasks are tracked and disagreements are resolved.
- Modeling Lead: Owns core MATLAB models/simulations

- Analysis/Validation Lead: Performs hand calculations and cross-checks simulation outputs
- Documentation & Visualization Lead: Compiles report (with figures/tables), prepares slides (if needed), and builds visual artifacts (e.g. animations or plots)
- Quality Assurance & Reproducibility Lead: Enforces coherent style across artifacts, ensures work is not overwritten across different versions, ensures data quality, tests/re-runs codes to ensure robustness and reproducibility

3) Communication Norms

- Primary channel: Discord
- Secondary channel: Text
- Response time: within 24 hours on weekdays and 48 hours on weekends
- Meeting cadence: Weekly working meeting 60-90 minutes on Discord on Wednesdays 7pm
- Decision records: PM logs major decisions in a running document

4) Collaboration & Tools

- Version control: One shared Github Repo (e.g. one shared GitHub repo or Google Drive folder)
- Data management: All datasets, parameters, and assumptions documented in README.md

File naming & style: e.g. Snake_case, descriptive names (add date)

- Backups: Date in file name then sort later (e.g. PM tags old versions according to numbered system)

5) Quality Standards (Definition of 'Done')

1. At least two drone design sketches showing key geometric features
2. Results of thrust-to-weight analysis presented either in a summary table or plot
3. Results of finite element analysis
 - Numerical values presented in a summary table
 - Visualizations of x-, y-, and z-displacement and Von Mises stress
4. A final design recommendation identified using the results of research on

Requirements meet project description, code is reproducible (runs from the README.md file, all specified figures regenerate, at least one teammate has tested edge cases, all assumptions, units, and references are documented in report)

6) Timeline & Milestones

- Week 1: Problem framing, literature/background review, dataset/parameters
- Week 2: First pass models (hand calculations + baseline MATLAB scripts); initial plots/visuals
- Week 3: Validation and quality assurance
- Week 4: Report, slides, presentation rehearsal, repository cleanup

7) Decision-Making & Conflict Resolution

- Default: consensus after discussion; PM facilitates
- If blocked >48 hrs: Time-limited debate, majority vote, PM documents decision
- Technical disagreements: Evidence-based comparison before vote
- Escalation: If unresolved, consult instructor/mentor with a concise memo.

8) Workload, Accountability & Attendance

- Task board: To-Do / Doing / Review / Done with owners and due dates documented in README.md)
- Accountability: Missed deadline → recovery plan within 24 hrs; repeated misses trigger role redistribution.
- Attendance: If you'll miss a meeting, give 24-hour notice and share updates asynchronously.

10) Accessibility & Inclusion

We commit to respectful collaboration, equitable turn-taking, and flexibility for documented accessibility needs.

11) Deliverables Checklist

Code repository with clear structure and README, hand calculations linked to modeled results for cross-validation, specific figures & visualizations, final report with specified sections.

12) Agreement & Signatures

Name & Signature: Nathan Huynh Date: 07/08/2026

Name & Signature: Annie Dang Date: 07/08/2026

Name & Signature: Bao Nguyen Date: 07/08/2026

Name & Signature: Jonah Williamson Date: 07/08/2026